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Quantification of the redundancy distribution in truss and beam structures

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ABSTRACT

The degree of statical indeterminacy as a fundamental property in structural mechanics is today mainly known as a property of a complete system without any information about its spatial distribution. The redundancy matrix provides information about the distribution of statical indeterminacy in the system and by this gives an additional valuable insight into the load-bearing behaviour. The derivation and definition of the redundancy matrix are presented based on truss systems and its mathematical properties and their mechanical interpretations are provided. The definition of the redundancy matrix is extended to other discrete systems like beam structures and a definition of the redundancy density is given for the continuous 1D case. Potential applications of the concept include robust design of structures, quantification of imperfection sensitivity as well as assessment of optimal actuator placement in adaptive structures.

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1. Introduction

1.1. Motivation

The degree of statical indeterminacy is a fundamental property in structural mechanics. It is an important indicator for the load-bearing behaviour and the structure's reaction to constraints like temperature or support settlement. For statically indeterminate systems the equilibrium conditions are not sufficient to determine all internal forces and reaction forces. The internal forces depend on the stiffness distribution and therefore the load transfer can be manipulated by modifying material or cross sectional properties ("stiffness attracts forces"). By increasing the degree of statical indeterminacy not only the stiffness but also the redundancy, i.e. the safety and ductility of a structure, can be increased. In contrast, in statically determinate systems constraint load cases do not cause any internal forces.

The degree of statical indeterminacy is defined as the difference between the number of unknown internal forces and the number of available equations of equilibrium. It can be calculated by formulas dating back to Maxwell (1864) based on the number of joints and members and kinematic constraints. Pellegrino and Calladine (1986) describe a more comprehensive approach which is based on the rank of the equilibrium matrix and makes use of the four fundamental vector subspaces associated with the equilibrium matrix. Another approach to determine the statical indeterminacy that focuses more on the mechanical understanding of the structure is the decomposition into statically determinate basic structures. All approaches provide the degree of indeterminacy of the complete system as one unique global information but they provide little information about its spatial distribution.

The main purpose of this paper is to introduce the redundancy matrix as a basis for additional insight into structural properties beyond the classical definition of statical indeterminacy. The analogy between the redundancy matrix and the distribution of the statical indeterminacy in the structure opens a wide field for further investigations. The concept has the potential to extend assessment of redundancy in structures beyond the insight provided by the degree of statical indeterminacy. For instance in the geometrically non-linear case, the analogy does not hold any more. Although the notion of statical indeterminacy is known for many decades or even centuries, there are still interesting things to discover.

1.2. State of the Art

The analogy between adjustment calculus in geodesy and structural systems is known already since a long time and was described e.g. by Fischer (1899) and Finsterwalder (1903) in the early 20th century. Cappilleri (1911) and Basch (1913) use graphical methods known from statics to solve least squares equaliza-

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tion problems and Passer (1933) describes the analogy between the compensation for line networks and constraints in statically indeterminate truss systems.

Linkwitz (1961) revisited this analogy and describes it in a matrix notation. He already defines an idempotent matrix, similar to the redundancy matrix, but has not yet recognized its mechanical interpretation and significance. Bahndorf (1991) and Ströbel (1995) from the group of Prof. Klaus Linkwitz in Stuttgart followed his ideas and first presented the concept of elastic redundancy, which provides information about the distribution of statical indeterminacy in the structure. Eriksson and Tibert (Eriksson and Tibert, 2006; Tibert, 2005) used this idea to study the sensitivity to manufacturing imperfections of a tension truss antenna.

In Ströbel and Singer (2008) the redundancy or distributed statical indeterminacy is used to estimated the imperfection sensitivity for the production of textile membranes and inflatable structures represented by grid systems. They also proposed a first basic approach to apply the redundancy for the optimization of truss structures by eliminating the element with the largest redundancy number step by step.

In the context of reanalysis of discrete structures, Kolakowski et al. (2008) used a so-called "strain influence matrix". The rank of this matrix was recognized to be equal to the degree of statical indeterminacy, however without describing the information on its spatial distribution.

Zhou et al. (2015) propose a method for the calculation of the distributed statical indeterminacy that is also applicable to kinematically indeterminate assemblies, i.e. structures with kinematic modes, and introduces also the distributed kinematic indeterminacy. Chen et al. (2018) showed that this approach is identical to the original formulation by Ströbel (1995) if the generalized inverse theory is introduced to avoid the limitation of kinematic structures with singular stiffness matrices. In addition, they proposed a unified method for symmetry representations and distributed indeterminacy evaluation of tensegrity structures by introducing group representation theory and full symmetry subspaces.

1.3. Applications

Information about the distribution of statical indeterminacy or redundancy inside a structure can help designing robust structures and provides a deeper understanding of the load-bearing behaviour. Determinate substructures or highly redundant areas of a structure can be easily identified.

In addition to the mere understanding of the load-bearing behaviour there are several areas of application for the redundancy or indeterminacy distribution. It directly provides information about the imperfection sensitivity, it can be used for structural optimization or to define a robustness index (Kou et al., 2017). The redundancy matrix can also be helpful in the design of adaptive structures (Wagner et al., 2018) or the analysis of shape and stress control problems (Kawaguchi et al., 1996).

The present study summarizes the developments in the field of redundancy matrices during the last decades and extends its definition from truss systems to frame structures and a one-dimensional continuum. The application of available techniques for calculating the redundancy distribution in truss structures with one deformation mode per element to frame systems with several deformation modes per element yields straightforward results. Whereas the extension to continuous systems in 2D or 3D is more challenging and breaking new ground. It could provide valuable inside also into the load-bearing behaviour of continuous systems. Structural theories that are internally statical indeterminate and non-linear or dynamic analyses are further challenges that could lead to a new and broader understanding of statical indeterminacy. This goes beyond current textbook knowledge of structural mechanics and might open up huge unexploited potentials with far reaching consequences for the mechanical understanding of structures. It is briefly shown, how the redundancy matrix can be applied to the design of adaptive structure and robust design optimization.

1.4. Outline

In Section 2 the redundancy matrix and distributed statical indeterminacy will be first introduced for discrete truss systems, following the definition of Ströbel (1995). After that, mechanical and mathematical properties of the redundancy matrix will be discussed in Section 3. In the last two sections, the idea of distributed redundancy will be extended to plane beam problems in Section 4 and to the continuous 1D case in Section 5, introducing the notion redundancy density in structures.

2. Derivation and definition of redundancy matrix

The redundancy distribution describes the constraint of the surrounding structure on every member. It can be calculated as the response of a member to a prescribed unit elongation. The redundancy r of an element in a truss system is defined as

$$r = \frac{\Delta l_0 - \Delta l}{\Delta l_0} = \frac{-\Delta l_{\mathrm{el}}}{\Delta l_0} \tag{1}$$

where Δl_0 is a prescribed initial elongation of the unconstrained member before assembly, Δl is the resulting total elongation and $\Delta l_{\mathrm{el}} = \Delta l - \Delta l_0$ is the elastic part of the elongation of the element. Technically, Δl_0 can be interpreted as a fabrication imperfection or an elongation due to temperature change. In the following, this definition will be applied to a matrix formulation of a discrete truss system.

For discrete truss structures, the governing equations describing static equilibrium can be directly written and solved in discrete form. The equilibrium equations describe the relation of n_{d} external forces at the nodes in the vector $\hat{\mathbf{b}}$ and the vector $\mathbf{N}$ of internal normal forces in n_{e} elements by the $n_{\mathrm{d}} \times n_{\mathrm{e}}$ equilibrium matrix $\mathbf{A}^{\mathrm{T}}$.

$$\mathbf{A}^{\mathrm{T}}\mathbf{N} = \hat{\mathbf{b}} \tag{2}$$

The internal normal forces of all elements are related to the elongations Δl of the elements by the member stiffnesses on the diagonal entries of the $n_{\mathrm{e}} \times n_{\mathrm{e}}$ member stiffness matrix $\mathbf{C}$. For truss elements with only one deformation mode (excluding the rigid body modes) these member stiffnesses are EA/l.

$$\mathbf{N} = \mathbf{C}\Delta\mathbf{l}_{\mathrm{el}} \qquad \Delta\mathbf{l}_{\mathrm{el}} = \Delta\mathbf{l} - \Delta\mathbf{l}_0 \tag{3}$$

The elastic elongation Δl_{el} is defined as the difference of the total elongation Δl and an initial prescribed elongation Δl_0, see Eq. (1). The total elongations are calculated from the set of n_{d} unconstrained nodal displacements in vector $\mathbf{d}$ using the $n_{\mathrm{e}} \times n_{\mathrm{d}}$ compatibility matrix $\mathbf{A}$, which is the transpose of the equilibrium matrix $\mathbf{A}^{\mathrm{T}}$.

$$\Delta\mathbf{l} = \mathbf{A}\mathbf{d} \tag{4}$$

Substituting the internal forces and element elongations, all three Eqs. (2)–(4) can be combined to a direct relation of nodal forces and nodal displacements,

$$\left(\mathbf{A}^{\mathrm{T}}\mathbf{C}\mathbf{A}\right)\mathbf{d} - \mathbf{A}^{\mathrm{T}}\mathbf{C}\Delta\mathbf{l}_0 = \hat{\mathbf{b}} \tag{5}$$

which can be solved for the unknown nodal displacements.

$$\mathbf{d} = \left(\mathbf{A}^{\mathrm{T}}\mathbf{C}\mathbf{A}\right)^{-1}\left[\hat{\mathbf{b}} + \mathbf{A}^{\mathrm{T}}\mathbf{C}\Delta\mathbf{l}_0\right] \tag{6}$$

$\left(\mathbf{A}^{\mathrm{T}}\mathbf{C}\mathbf{A}\right) = \mathbf{K}$ is the global stiffness matrix, which could also be obtained, for example, by the direct stiffness method.

As the redundancy distribution is a property of the structure and independent of external loads, only the part that is independent of $\hat{\mathbf{b}}$ is considered. Using the compatibility matrix $\mathbf{A}$, from the nodal displacements $\mathbf{d}$ the resulting total element elongations can be calculated (Eqs. (4)

$$\Delta\mathbf{l} = \underbrace{\left[\mathbf{A}\left(\mathbf{A}^{\mathrm{T}}\mathbf{C}\mathbf{A}\right)^{-1}\mathbf{A}^{\mathrm{T}}\mathbf{C}\right]}_{\mathbf{A}_0}\Delta\mathbf{l}_0 \qquad \mathbf{A}_0 = \left[\mathbf{A}\left(\mathbf{A}^{\mathrm{T}}\mathbf{C}\mathbf{A}\right)^{-1}\mathbf{A}^{\mathrm{T}}\mathbf{C}\right] \tag{7}$$

To simplify notation in the sequel the total self-strain matrix $\mathbf{A}_0$ is introduced. It maps the initial elongations $\Delta\mathbf{l}_0$ to the total elongations $\Delta\mathbf{l}$. Similarly, also the elastic elongations can be computed using the identity matrix $\mathbf{I}$

$$\Delta\mathbf{l}_{\mathrm{el}} = \Delta\mathbf{l} - \Delta\mathbf{l}_0 = \underbrace{\left[\mathbf{A}_0 - \mathbf{I}\right]}_{-\mathbf{R}}\Delta\mathbf{l}_0 \qquad \mathbf{R} = \mathbf{I} - \mathbf{A}_0 \tag{8}$$

The redundancy matrix $\mathbf{R}$ maps the initial elongations $\Delta\mathbf{l}_0$ to the negative elastic elongations $\Delta\mathbf{l}_{\mathrm{el}}$.

A related equation was already used by Kawaguchi et al. (1996) for shape and stress control analysis. There, the normal forces $\mathbf{N}$ are derived from the initial elongations $\Delta\mathbf{l}_0$ by the lower part $\mathbf{G}_{\mathrm{l}}$ of a matrix $\mathbf{G}$.

$$\mathbf{N} = \mathbf{G}_{\mathrm{l}}\Delta\mathbf{l}_0 \tag{9}$$

Multiplying this equation by the matrix of member flexibilities $\mathbf{F} = \mathbf{C}^{-1}$ transfers it into a relation between the initial elongations $\Delta\mathbf{l}_0$ and the elastic elongations $\Delta\mathbf{l}_{\mathrm{el}}$

$$\Delta\mathbf{l}_{\mathrm{el}} = \mathbf{F}\mathbf{N} = \mathbf{F}\mathbf{G}\Delta\mathbf{l}_0 \tag{10}$$

Comparing it to Eq. (8) $-\mathbf{F}\mathbf{G}_{\mathrm{l}}$ can formally be identified as the redundancy matrix $\mathbf{R}$. However, this modification leading to the redundancy matrix was not exploited in Kawaguchi et al. (1996). It is the purpose of the present study to directly derive the redundancy matrix and discuss its specific properties and the relation to the distributed statical indeterminacy.

For kinematically indeterminate structures Eq. (7) cannot be evaluated, because the stiffness matrix $\mathbf{K} = \left(\mathbf{A}^{\mathrm{T}}\mathbf{C}\mathbf{A}\right)$ is singular. In these cases the generalized inverse of the stiffness matrix $\mathbf{K}$ can be used to avoid this limitation (Chen et al., 2018). As the loads are stemming from element elongations $\mathbf{A}^{\mathrm{T}}\mathbf{C}\Delta\mathbf{l}_0$ only, they are not in the null space of the stiffness matrix $\mathbf{K}$; this application of the generalized inverse does not lead to any difficulties.

Therefore, the presented approach also works for systems that are statically indeterminate in one direction or mode and are kinematically indeterminate in another direction, like in the example presented in Pellegrino (1993). A system of two truss elements arranged in a straight line supported at both outer ends will be classified as statically determinate by methods only using the number of nodes, elements and supports. The redundancy matrix will recognise the statical indeterminacy in longitudinal direction, even though the system is kinematically indeterminate in transverse direction.

As a first example, the redundancy calculation will be demonstrated for the simplest truss structure with a statical indeterminacy of 1. The system consists of three truss elements with a constant Young's modulus E and cross sectional area A for all elements and two degrees of freedom not constrained by a boundary condition, as shown in Fig. 1(left).

According to the definition of the redundancy of an individual element in Eq. (1) a unit elongation $\Delta l_{0,1} = 1$ is prescribed for the first element. This prescribed elongation is equivalent to a force $f_x = 1/5EA$ in horizontal direction acting on node 1 and leads to a displacement

$$\mathbf{u} = \begin{bmatrix} 0.793 \\ -0.207 \end{bmatrix} \tag{11}$$

of this node. The resulting total elongation of element 1 is $\Delta l_1 = 0.793$ and the redundancy for this element is obtained as

$$r = \frac{1.0 - 0.793}{1.0} = 0.207 \tag{12}$$

For the calculation of the redundancy distribution in matrix notation according to Eqs. (7) and (8) the compatibility matrix $\mathbf{A}$ and the member stiffness matrix $\mathbf{C}$ are required. In this example, the compatibility matrix $\mathbf{A}$ describes the relation of the two degrees of freedom $\mathbf{d}$ at node 1 and the three element elongations $\Delta\mathbf{l}$.

$$\underbrace{\begin{bmatrix} \Delta l_1 \\ \Delta l_2 \\ \Delta l_3 \end{bmatrix}}_{\Delta\mathbf{l}} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} d_1 \\ d_2 \end{bmatrix}}_{\mathbf{d}} \tag{13}$$

If we assume the same Young's modulus E and cross sectional area A for all three elements we can use the member stiffnesses in symbolic form in the matrix $\mathbf{C}$ according to Eq. (3), as the redundancy distribution only depends on the ratio of stiffnesses but not on the absolute values.

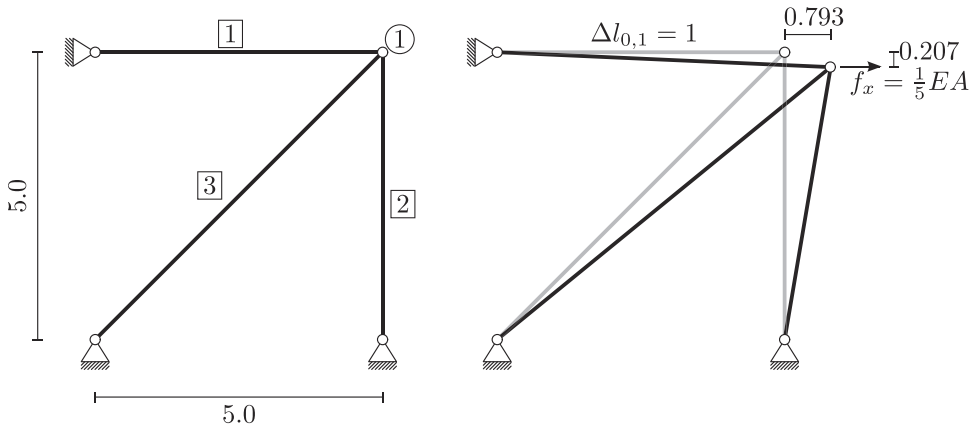


Fig. 1. System (left) and redundancy calculation for element 1 (right) for a model truss system.

$$\mathbf{C} = \begin{bmatrix} \frac{EA}{L} & 0 & 0 \\ 0 & \frac{EA}{L} & 0 \\ 0 & 0 & \frac{EA}{\sqrt{2}L} \end{bmatrix} \tag{14}$$

From this, the redundancy matrix for the system can be calculated by Eqs. (7) and (8) as

$$\mathbf{R} = \begin{bmatrix} 0.207 & 0.207 & -0.293 \\ 0.207 & 0.207 & -0.293 \\ -0.414 & -0.414 & 0.586 \end{bmatrix} \tag{15}$$

3. Mathematical and mechanical properties of the redundancy matrix

The degree of statical indeterminacy n_s for truss structures in 2D can be calculated from the number of elements n_e, the number of nodes n_n and the number of support constraints n_c.

$$n_s = n_e - (2n_n - n_c) = n_e - n_d, \tag{16}$$

where $n_d = 2n_n - n_c$ is the number of unconstrained degrees of freedom. If $n_e = n_d$, the degree of statical indeterminacy is zero and the structure is nominally statically determinate. For certain geometrical configurations, the system might still contain a pathological kinematic mechanism. A von Mises truss with all three nodes on a straight line, is statically indeterminate in the direction of the line and kinematic orthogonal to it. Classical formulas for the degree of statical indeterminacy, like Eq. (16), do not recognise these kinematic mechanisms. Whereas during the calculation of the redundancy matrix they can be identified due to the singularity of the stiffness matrix. The equilibrium matrix $\mathbf{A}^T$ becomes a square matrix of size $n_e \times n_e$ and the internal forces can be determined directly from equilibrium. For statically indeterminate structures $\mathbf{A}^T$ will be a rectangular matrix of size $n_d \times n_e$ and rank n_d.

The trace of the total self-strain matrix $\mathbf{A}_0$ and redundancy matrix $\mathbf{R}$ can be calculated as n_d and n_s respectively.

$$\mathrm{tr}(\mathbf{A}_0) = \mathrm{tr}\left(\mathbf{A}\left(\mathbf{A}^T\mathbf{C}\mathbf{A}\right)^{-1}\mathbf{A}^T\mathbf{C}\right) = \mathrm{tr}\left(\left(\mathbf{A}^T\mathbf{C}\mathbf{A}\right)^{-1}\mathbf{A}^T\mathbf{C}\mathbf{A}\right) = \mathrm{tr}(\mathbf{I}_{n_d}) = n_d \tag{17}$$

$$\mathrm{tr}(\mathbf{R}) = \mathrm{tr}(\mathbf{I}_{n_e}) - \mathrm{tr}(\mathbf{A}_0) = n_e - n_d = n_s \tag{18}$$

The trace of the redundancy matrix $\mathbf{R}$ is equal to the degree of statical indeterminacy n_s and therefore the diagonal entries R_{ii} of the redundancy matrix can be interpreted as the contributions of the individual elements to the statical indeterminacy n_s. The diagonal entries are also called *distributed statical indeterminacy* (Chen et al., 2018; Zhou et al., 2015; Eriksson and Tibert, 2006).

If an element i is fixed between two rigid supports or surrounded by an infinitely stiff structure then $R_{ii} = 1$. This element has a high redundancy, i.e. it could be removed from the structure without altering the load bearing behaviour. On the contrary, if an element is free to elongate $R_{ii} = 0$. The redundancy for this element is zero, i.e. the structure will (partially) fail if this element is removed. It is part of a statically determinate structure or substructure.

The off-diagonal entries in column i in the redundancy matrix are an indicator for the influence of an initial elongation in element i on the surrounding structure. The number of non-zero entries in this column indicates the number of elements affected and the value is an indicator for the magnitude of the normal force induced.

The entries of the redundancy matrix are a direct measure for the imperfection sensitivity of a structure and its individual members. A small diagonal value in the redundancy matrix indicates that a fabrication imperfection for this member will lead only to small constraints and therefore only to small forces in the structure. In the context of adaptive structures, an actuator in this element could be used for manipulation of the deformation with little constraints. Vice versa, an actuator in a member with a high diagonal value in the redundancy matrix will cause larger forces and can be used to manipulate the force distribution in the structure.

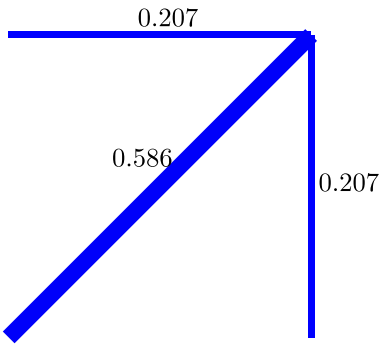


Fig. 2. Redundancy distribution for a didactic truss system.

For the didactic example introduced in Section 2, the distribution of statical indeterminacy is visualized in Fig. 2. The degree of statical indeterminacy for the system $n_S = 1$ is equal to the sum of these diagonal entries $n_S = \mathrm{tr}(\mathbf{R})$.

Of course, the redundancy distribution only brings a real advantage in understanding the load-bearing behaviour with larger and more complex structures. In Fig. 3 the redundancy distribution in a spatial truss system with 288 elements with constant EA is shown. The four highlighted nodes are supported in all three spatial directions. Red indicates low redundancy and blue indicates high redundancy. The total degree of statical indeterminacy is equal to 45.

It can be seen that redundancy at the edge is in general lower, especially in the corners. Here, the redundancy in all three elements connected to the corner node is equal to zero. These small subsystem are statically determinate. The bars in the upper and lower level in the centre have the highest redundancy. It is also higher than the redundancy in the diagonal elements.

Due to the structure of the matrix products in the definition of the elongation matrix $\mathbf{A}_0$, both the elongation matrix $\mathbf{A}_0$ and the redundancy matrix $\mathbf{R}$ are idempotent matrices, i.e. $\mathbf{RR} = \mathbf{R}$. An idempotent matrix is always diagonalizable and its eigenvalues are either 0 or 1. The trace of an idempotent matrix equals the rank of the matrix.

The matrices $\mathbf{A}_0$ and $\mathbf{R}$ share the same eigenvectors $\boldsymbol{\phi}$

$$\begin{aligned} \mathbf{R}\boldsymbol{\phi} &= \lambda\boldsymbol{\phi} \\ (\mathbf{I} - \mathbf{A}_0)\boldsymbol{\phi} &= \lambda\boldsymbol{\phi} \\ \mathbf{A}_0\boldsymbol{\phi} &= \mathbf{I}\boldsymbol{\phi} - \lambda\boldsymbol{\phi} \\ \mathbf{A}_0\boldsymbol{\phi} &= (1-\lambda)\boldsymbol{\phi} \end{aligned} \tag{19}$$

but the eigenvalues for these eigenvectors are 0 and 1 for $\mathbf{A}_0$ and or 1 and 0 for $\mathbf{R}$. This means, that the image (im) of the redundancy matrix is the kernel (ker) of the total self-strain matrix and vice versa:

$$\mathrm{im}(\mathbf{R}) = \ker(\mathbf{A}_0) \quad \ker(\mathbf{R}) = \mathrm{im}(\mathbf{A}_0) \tag{20}$$

The image of the redundancy matrix $\mathrm{im}(\mathbf{R})$ spans a n_s dimensional space of purely incompatible elongations in a structure. Initial elongations from this subspace will lead to no deformation of the structure as $\Delta\mathbf{l} = \boldsymbol{0}$ but to a stress state that is in self-equilibrium. The subspace is equivalent to the space of self-stress states described e.g. by Pellegrino and Calladine (1986) and Pellegrino (1993). On the contrary, the image of the elongation

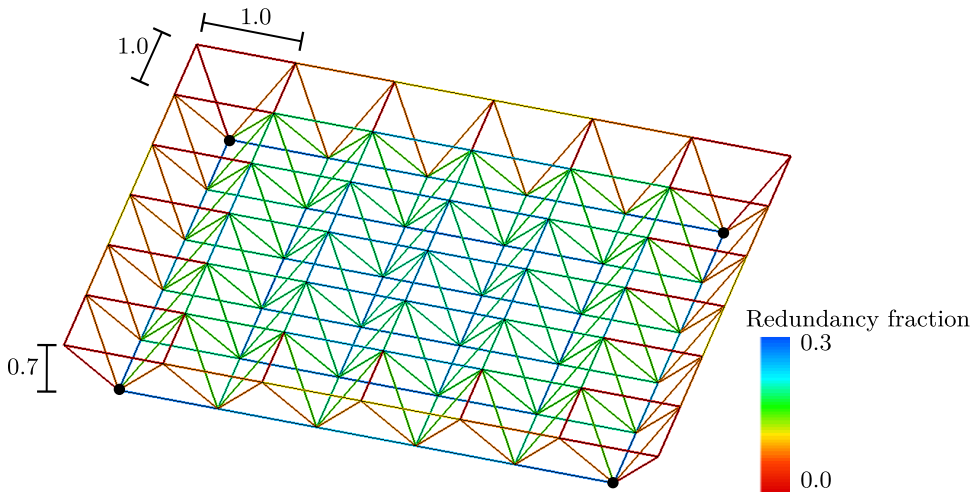


Fig. 3. Redundancy distribution in a spatial truss system.

matrix $\mathrm{im}(\mathbf{A}_0)$ spans the n_d dimensional space of compatible elongations, i.e. initial elongations that lead only to deformations but yield no internal forces, as $\Delta\mathbf{l}_\mathrm{el} = \boldsymbol{0}$.

The redundancy matrix and total self-strain matrix can also be interpreted as projections. The redundancy matrix $\mathbf{R}$ describes a parallel projection of initial elongations into the subspace of elastic elongations ($\mathrm{im}(\mathbf{R})$) parallel to the subspace of compatible elongations ($\ker(\mathbf{R})$), compare Eq. (8). The total self-strain matrix $\mathbf{A}_0$ describes exactly the opposite parallel projection, the projection of initial elongations into the subspace of compatible elongations ($\mathrm{im}(\mathbf{A}_0)$), compare Eq. (7).

For the redundancy matrix and total self-strain matrix of the didactic example introduced in Section 2 the eigenvalues and eigenvectors can be calculated as follows:

$$\boldsymbol{\lambda}_\mathrm{R} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \boldsymbol{\Phi} = \begin{bmatrix} -0.5 & \sqrt{2} & -1.0 \\ -0.5 & 0 & 1.0 \\ 1 & 1 & 0 \end{bmatrix} \quad \boldsymbol{\lambda}_{\mathrm{A}_0} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \tag{21}$$

Using these eigenvectors as prescribed elongations will lead either to purely elastic elongations with no deformations or to zero elastic deformations, i.e. deformation with no internal forces.

The redundancy matrix and total self-strain matrix of the didactic example are given in Eq. (21). Applying the first eigenvector ϕ_1 as prescribed elongations to the structure will yield purely elastic deformations in the system, as this vector is part of the image of $\mathbf{R}$. The system remains undeformed as the total elongations $\Delta\mathbf{l}$ are zero (Fig. 4, left column). The prescribed elongations completely become elastic deformations equivalent to the self-stress state of the system.

The second and third eigenvectors ϕ_2 and ϕ_3 are part of the kernel of $\mathbf{R}$. Prescribing these elongations will yield only total elongations and zero elastic deformations. These are compatible deformations with no internal forces (Fig. 4, centre and right column). Any vector of prescribed elongations that is not completely part of the image or kernel of $\mathbf{R}$ will yield elastic and total deformations. The projection matrices $\mathbf{R}$ and $\mathbf{A}_0$ split the prescribed elongations into the self-stress and compatible parts by Eqs. (7) and (8).

4. Redundancy for beams and frames

The definition of the redundancy matrix described in Section 2 can be straightforwardly extended to beam elements. In Ströbel (1995) the diagonal entries of the redundancy matrix are also shown for a small frame structure, but no details of the derivation or computation are given. In the context of structural system vulnerability and statistical theory Jiang et al. (2012) introduce a redundancy matrix for frame structures. It is based on the equation of a redundant restriction matrix and contains redundancies not only for the members, but also for the joints. It is therefore a different approach.

wFor plane Euler–Bernoulli and Timoshenko beams there are three deformation modes and therefore also three internal forces and strain-like variables instead of one for the truss element described in Section 2. Different choices for the set of three deformation modes are possible. The choice of one specific set of deformation modes will define the compatibility matrix $\mathbf{A}$ and the stiffnesses of these modes in the member stiffness matrix $\mathbf{C}$. This reduction to the non-rigid-body modes is well known in the classical theory of matrix structural analysis and described in many text books (e.g. Livesley, 1964; Przemieniecki, 1968).

Not all sets of deformation modes lead to a consistent definition of the redundancy matrix. In many cases, the redundancy matrix will not be independent of rigid body rotations. To avoid these dependencies, deformation modes based on an eigenvector decomposition of the stiffness matrix $\mathbf{k}$ of a single beam element are chosen. The normalization of the eigenvectors is transferred to the eigenvalues by dividing each eigenvalue by the square of the length of the corresponding eigenvector. The scaled eigenvalues $\tilde{\boldsymbol{\lambda}}$ and original eigenvectors $\boldsymbol{\phi}$ are also a valid decomposition of the element stiffness matrix $\mathbf{k} = \boldsymbol{\phi}\tilde{\boldsymbol{\lambda}}\boldsymbol{\phi}^\mathrm{T}$.

For the plane Euler–Bernoulli beam they are given as

$$\tilde{\boldsymbol{\lambda}} = \begin{bmatrix} \frac{EA}{l} & 0 & 0 \\ 0 & \frac{3EI}{l} & 0 \\ 0 & 0 & \frac{EI}{l} \end{bmatrix} \quad \boldsymbol{\phi} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & \frac{-2}{l} & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 0 \\ 0 & \frac{2}{l} & 0 \\ 0 & 1 & 1 \end{bmatrix} \tag{22}$$

For plane Timoshenko beams the same eigenvectors $\boldsymbol{\phi}$ can be used, only the second eigenvalue $\tilde{\lambda}$ now includes the shear stiffness GA

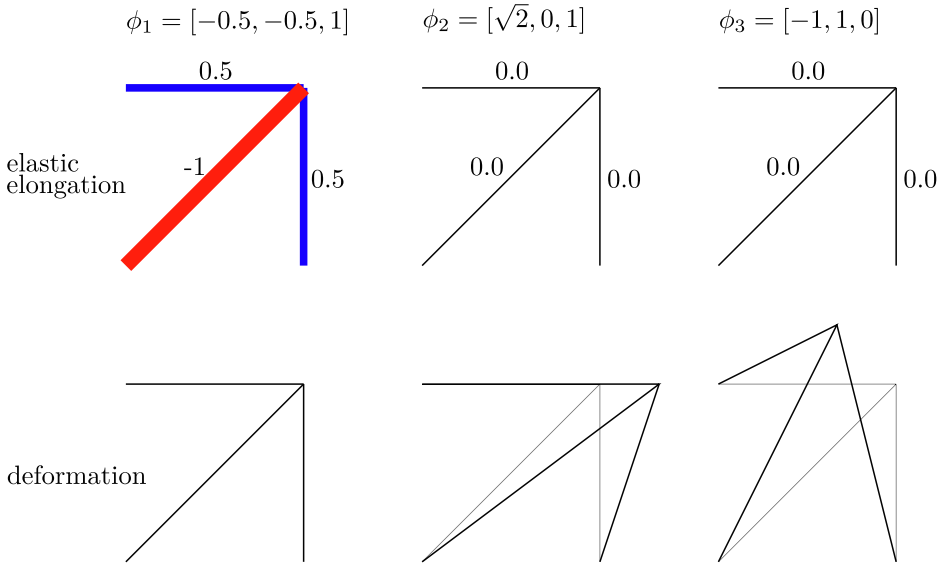


Fig. 4. Elastic deformation (top) and deformation (bottom) of an didactic truss system due to prescribed elongations distributed like the eigenvectors of the redundancy matrix.

$$\tilde{\boldsymbol{\lambda}} = \begin{bmatrix} \frac{EA}{l} & 0 & 0 \\ 0 & \frac{3lGAEI}{GAl^2+12EI} & 0 \\ 0 & 0 & \frac{EI}{l} \end{bmatrix} \qquad \boldsymbol{\phi} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & \frac{-2}{l} & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 0 \\ 0 & \frac{2}{l} & 0 \\ 0 & 1 & 1 \end{bmatrix} \tag{23}$$

For both types of beam elements, the six rows of the matrix of eigenvectors $\boldsymbol{\phi}$ correspond to the six degrees of freedom: axial displacement u_1, transverse displacement w_1 and rotation φ_1 at the first node and axial displacement u_2, transverse displacement w_2 and rotation φ_2 at the second node.

Applying the scaling to unit length to the eigenvectors directly and not to the eigenvalues due to the different physical units of displacements and rotations the result for the redundancy matrix will depend on the chosen unit for the length dimension.

Due to the three deformation modes, one will obtain three diagonal entries in the redundancy matrix per element. These describe the redundancy or statical indeterminacy of one element in the specific deformation mode. Each of these three values can vary between zero and one. Therefore, the total degree of statical indeterminacy of one element can be up to three: one for the elongation and two for the combination of shear and bending. Due to the coupling of shear and bending, the redundancy for the bending and shear modes will often be added together in the results of the following analyses.

In the following example, we show the results for a frame structure with Euler–Bernoulli beam elements and discuss the relation of the redundancies in the elongation and bending modes. The structure consists of a square frame clamped at both sides, equipped with a diagonal bracing with hinges at both ends (Fig. 5). The degree of statical indeterminacy for the entire system is equal to four.

The redundancy matrix for this example has 12×12 entries, corresponding to 4 elements with 3 deformation modes each (one elongation mode and two bending modes) and is shown in Eq. (24).

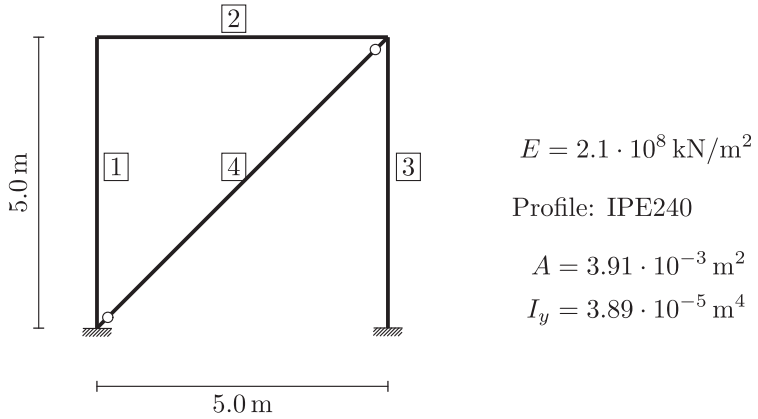


Fig. 5. System for a frame structure example.

$$
\mathbf{R} = \left[\begin{array}{ccc|ccc|ccc|ccc}
\mathbf{0.0019} & -0.0035 & -0.0012 & 0.0014 & 0.0047 & 0.0000 & 0.0009 & -0.0035 & -0.0012 & -0.0040 & 0.0 & 0.0 \\
-0.1171 & \mathbf{0.5907} & -0.1318 & -0.2363 & -0.2927 & 0.1662 & -0.1560 & 0.0921 & 0.0344 & 0.3862 & 0.0 & 0.0 \\
-0.1212 & -0.3954 & \mathbf{0.8653} & 0.1582 & -0.3029 & 0.1670 & 0.2379 & 0.1036 & 0.0323 & -0.1650 & 0.0 & 0.0 \\
\hline
0.0014 & -0.0071 & 0.0016 & \mathbf{0.0028} & 0.0035 & -0.0020 & 0.0019 & -0.0011 & -0.0004 & -0.0046 & 0.0 & 0.0 \\
0.1576 & -0.2927 & -0.1010 & 0.1171 & \mathbf{0.3939} & 0.0002 & 0.0769 & -0.2934 & -0.1007 & -0.3315 & 0.0 & 0.0 \\
0.0003 & 0.4986 & 0.1670 & -0.1994 & 0.0007 & \mathbf{0.6663} & 0.0004 & -0.5002 & -0.1667 & -0.0009 & 0.0 & 0.0 \\
\hline
0.0009 & -0.0047 & 0.0024 & 0.0019 & 0.0023 & 0.0000 & \mathbf{0.0028} & -0.0047 & 0.0024 & -0.0053 & 0.0 & 0.0 \\
-0.1174 & 0.0921 & 0.0345 & -0.0368 & -0.2934 & -0.1667 & -0.1564 & \mathbf{0.5923} & -0.1322 & 0.3872 & 0.0 & 0.0 \\
-0.1209 & 0.1032 & 0.0323 & -0.0413 & -0.3022 & -0.1667 & 0.2382 & -0.3966 & \mathbf{0.8655} & -0.1660 & 0.0 & 0.0 \\
\hline
-0.0056 & 0.0163 & -0.0023 & -0.0065 & -0.0140 & 0.0000 & -0.0075 & 0.0163 & -0.0023 & \mathbf{0.0185} & 0.0 & 0.0 \\
0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & \mathbf{0.0} & 0.0 \\
0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & \mathbf{0.0}
\end{array}\right] \tag{24}
$$

Vertical and horizontal lines separate the contributions of each element. The diagonal elements corresponding to the distributed statical indeterminacy are printed in bold face.

The entries in the last two rows and columns are equal to zero, as element 4 is hinged at both ends and the two bending modes have no redundancy and are decoupled from the surrounding structure. Both the rank and the trace of the matrix are four, which is equal to the degree of statical indeterminacy.

The values of the diagonal elements of the redundancy matrix (Eq. (24)) are visualized in Fig. 6. For each element the redundancy in the elongation mode (first diagonal entry) and in the bending modes (sum of second and third diagonal entry) are shown. It can be seen that there is almost no redundancy in the elongation mode. The ratio of normal stiffness to bending stiffness $\frac{EAl^2}{EI} = 2512$ is relatively high and therefore the bending stiffness does not influence the elongation modes. Without bending, the structure behaves like a truss system and would be statically determinate with zero redundancy.

For the two bending modes the diagonal has exactly zero redundancy as it is decoupled due to the hinges. Both of these modes are not constrained by the surrounding structure.

The bracing element is of central importance for the structure. Removing it would have the largest influence on the load-bearing behaviour of the structure. This corresponds to the fact that it has the smallest total redundancy.

In contrast, adding for example a hinge in the outer frame would not alter the structural behaviour much. The two bending modes have a high redundancy and bending is not relevant for the load-bearing.

If the bending stiffness *EI* is increased by a factor of 1000, the elongation modes are also influenced by the larger bending stiffness and the truss-like behaviour for these modes does no longer exist. Half of the redundancy is now in the elongation modes (Fig. 7). The overall load-bearing behaviour is equally determined by elongation and bending action. The outer frame could carry the load also without the bracing element.

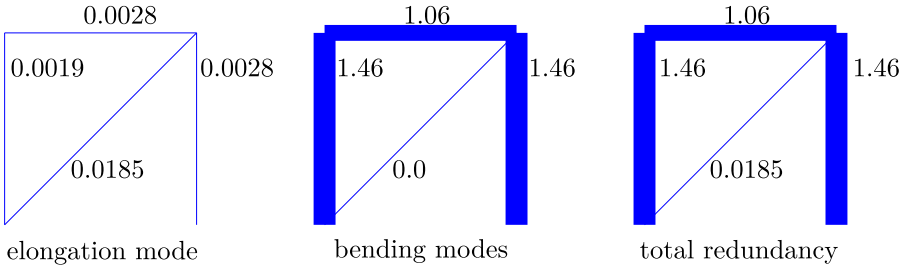


Fig. 6. Redundancy distributions for frame structure.

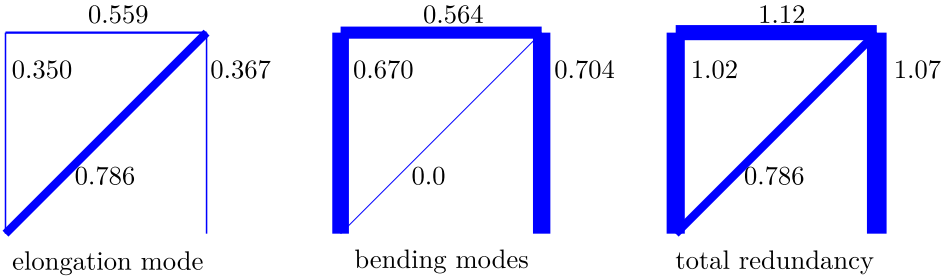


Fig. 7. Redundancy distributions for frame structure with 1000× larger bending stiffness *EI*.

An extension to spatial frame structures is straightforward.

5. Extension to continuous systems

The degree of statical indeterminacy and the redundancy distribution described so far are only defined for discrete systems of truss or beam elements. The application to continuous systems is an obvious extension of the approach. As a starting point, we describe the redundancy calculation for the continuous 1D case. The procedure for calculating the redundancy of a discrete truss element is transferred to an infinitesimal line element.

An initial elongation $\Delta l_0 = 1$ is applied to an infinitesimal line element of length $\mathrm{d}x$ at an arbitrary position x in the one-dimensional system, as shown in Fig. 8. For a constant EA this is equivalent to applying a pair of forces with the magnitude $EA/\mathrm{d}x$ at both ends of the infinitesimal line element. These forces correspond to the term $\mathbf{A}^{\mathrm{T}}\mathbf{C}\Delta\mathbf{l}_0$ in Eq. (6). The resulting displacements in the system can be calculated from the partial differential equation for the displacements $u(x)$ and the boundary conditions. The resulting total elongation of the infinitesimal line element is

$$\Delta l = \frac{l - \mathrm{d}x}{l}. \tag{25}$$

For the infinitesimal line element the definition of the redundancy in Eq. (1) can be rewritten as

$$\mathrm{d}R(x) = \frac{\Delta l_0 - \Delta l}{\Delta l_0} = \frac{1}{l}\mathrm{d}x = \bar{R}(x)\mathrm{d}x \tag{26}$$

where $\bar{R}(x)$ can be defined as a distributed redundancy or redundancy density.

From this, the total redundancy or statical indeterminacy of the system can be calculated by the integral of the redundancy density

$$R = \int_0^l \bar{R}(x)\mathrm{d}x = \int_0^l \frac{1}{l}\mathrm{d}x = 1 \tag{27}$$

The same procedure can also be applied to systems with non-constant stiffness distributions $EA(x)$. Here, the definition of the equivalent loads for an initial elongation $\Delta l_0 = 1$ becomes more complex and a series expansion is in many cases necessary to identify the redundancy density $\bar{R}(x)$.

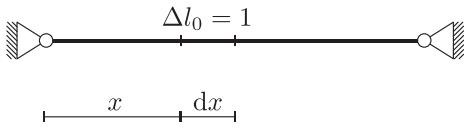


Fig. 8. One-dimensional continuous system.

For a linear stiffness distribution $EA(x) = 5 + 5x$ (Fig. 9 left) in the system shown in Fig. 8 the equivalent loads for an initial elongation $\Delta l_0 = 1$ of the infinitesimal line element of length $\mathrm{d}x$ at position x is

$$F = \frac{-5}{\ln(1+x) - \ln(1+x+\mathrm{d}x)} \tag{28}$$

resulting in a total elongation of the infinitesimal line element of

$$\Delta l = \frac{-\ln(x+\mathrm{d}x+1) + \ln(l+1) + \ln(x+1)}{\ln(l+1)}. \tag{29}$$

as the solution of the differential equation for the one-dimensional continuum.

From this $\mathrm{d}R(x)$ can be calculated as

$$\mathrm{d}R(x) = \frac{-\ln(x+1) + \ln(x+\mathrm{d}x+1)}{\ln(l+1)} \tag{30}$$

but no redundancy density can be identified explicitly. For this, a series expansion of $\mathrm{d}R(x)$ in $\mathrm{d}x$ is necessary.

$$\mathrm{d}R(x) \approx \frac{1}{(x+1)\ln(l+1)}\mathrm{d}x + \mathrm{HOT} \tag{31}$$

In this series higher order terms (HOT) of the infinitesimal line element $\mathrm{d}x$ can be neglected and the redundancy density can be identified as

$$\bar{R}(x) = \frac{1}{(x+1)\ln(l+1)} \tag{32}$$

The distribution of the redundancy density is shown for a structure of length 10 m in Fig. 9, right. The left part of the bar has a lower stiffness and therefore a higher redundancy density.

Also in this case, the total redundancy and also statical indeterminacy of the system can be calculated by the integral of the redundancy density

$$R = \int_0^l \bar{R}(x)\mathrm{d}x = \int_0^l \frac{1}{(x+1)\ln(l+1)}\mathrm{d}x = 1 \tag{33}$$

The redundancy density function shown in Fig. 9 and integrals of it provide the same information about the distribution of redundancy and statical indeterminacy in continuous structures as the redundancy matrix does for discrete structures. Therefore, it allows new insight into the load-bearing behaviour, imperfection sensitivity, robust design or actuation of continuous structures.

But more interestingly, the redundancy density function reveals that it is a special case of a influence line. When multiplied by the stiffness $EA(x)$ it corresponds to the normal force at the position of the loading caused by a dirac delta temperature load. This relation

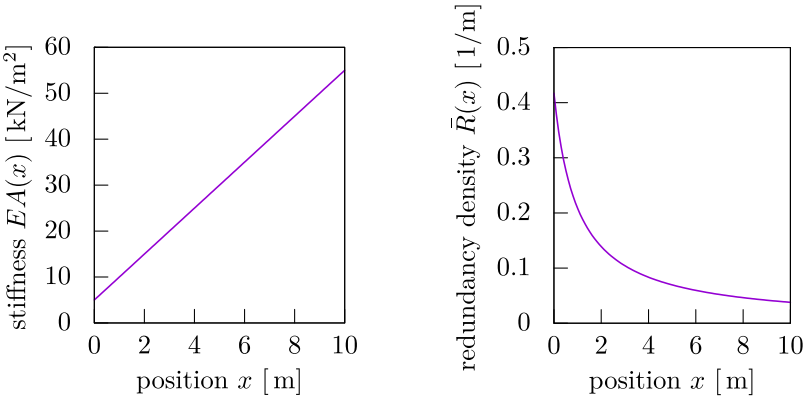


Fig. 9. Stiffness distribution and redundancy density for one-dimensional continuous system with linearly varying stiffness.

between the redundancy and influence lines is not evident from the redundancy matrix for the discrete case.

The extension of the concept of redundancy density to general solids and thin-walled structures is non-trivial and subject to current research.

6. Summary and outlook

The degree of statical indeterminacy as fundamental indicator for the load-bearing behaviour is a property of a structural system. However, this single number does not describe its distribution in the system. The redundancy matrix provides information about the distribution of the statical indeterminacy in the system and provides additional valuable insight into the load-bearing behaviour.

The developments in the field of the redundancy matrix during the past decades were summarized. The derivation and definition of the redundancy matrix were presented based on truss systems. The mathematical properties of the redundancy matrix and its mechanical interpretations were described. The diagonal entries are the distributed statical indeterminacy in a structure. The redundancy matrix and total self-strain matrix are projections of initial elongations onto the subspaces of elastic and compatible elongations, respectively.

The calculation of the redundancy matrix can be directly applied to other discrete systems like beam structures. Here, the choice of the correct set of deformation modes is essential to avoid length-scale dependencies and a coupling of the shear and bending modes and to obtain a consistent definition of the redundancy matrix. The dependency of the distribution of the statical indeterminacy on the ratio of normal stiffness to bending stiffness was demonstrated in a small example.

In order to use the distribution of the statical indeterminacy also for the assessment and design of two- or three-dimensional structures, an extension to continuous systems is necessary. The calculation of the redundancy density was described for the continuous 1D case and is consistent with the results for the discrete evaluation. The extension to 2D and 3D solids could provide valuable insight also into the load-bearing behaviour of continuous systems. Structural theories that are internally statically indeterminate and non-linear or dynamic analyses are further challenges that could lead to a new and broader understanding of statical indeterminacy and structural redundancy.

In addition to a deeper insight into the load-bearing behaviour provided by the distributed statical indeterminacy, the redundancy matrix and total self-strain matrix can also be used in other areas of structural mechanics. An obvious application is to use the redundancy distribution in the objective function of robust design optimization (Kou et al., 2017) or as measure for the imperfection sensitivity (Ströbel and Singer, 2008). Also for the design and analysis of adaptive structures, these matrices might provide useful information (Wagner et al., 2018). The redundancy matrix and total self-strain matrix describe the influence of an actuator on forces and deformations of other elements in a structure and could be used to design an efficient actuation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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